COLLECTION OF CTL DECISIONS

Standard: GENERALITY	**Sub clause: ---**	**Sheet n. 201A** **Page 1 (2)**
Subject: Characteristics of D.C. electric circuits for verifying normal operation and making and breaking capacity	**Key words:** **- D.C.** **- normal operation** **- making and breaking capacity**	**Decision n. 1 of doc. IECEE/CTL(SEC)019B/88, taken at 23rd meeting/1988**

To reach an acceptable test reproducibility, when verifying normal operation and making and breaking capacity, it is recommended that the test circuit has the following characteristics:

- **Voltage drop**

The maximum voltage drop which should not be exceeded, when passing from no-load to load condition, is 5% of the rated voltage.

- **Voltage ripple value**

The supply voltage ripple, at no-load condition, should not exceed 5%.

NOTES:

- Percentage ripple is given by r% = V_{ac}/V_{dc} x 100, where V_{ac} is the RMS value of the superimposed a.c. voltage and V_{dc} is the d.c. voltage.
A simple way to evaluate the ripple is to use an oscilloscope, whose input is selected in a.c. for the measurement of V_{ac} and in d.c. for the measurement of V_{DU}.

- The limitation of the ripple, which occurs, in practise, in rectifying converters, may be obtained choosing appropriately a rectifying bridge.

- The use of a three-phase rectifying bridge, which gives a theoretical ripple of about 4,2%, complies with this recommendation.

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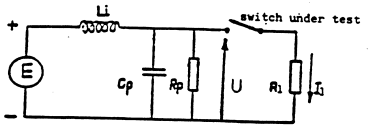
- **Recovery voltage**



Test circuit, where:
- E is the supply voltage at no-load condition;
- L_i is equivalent to the different self-inductances of the supply as well as the stray inductances of the rest of the test circuit;
- C_p is the stray capacitance of the circuit;
- R_p is the parallel damping resistance.

Neglecting the parallel damping resistor, R_p, the parameters L_i and C_p should be adjusted so as to obtain:

- $L_i = 2 \cdot 10^{-3}$ H ± 25%, and
- C_p = 50 μF ± 25%.

NOTE - Adopting the above mentioned parameters, all CTL Laboratories will get approximately

- the same time constant $T = L_i / R_i = 2 \cdot 10^{-3} / R_i$;

- the same oscillatory frequency $f_o = 1 / 2 \cdot (L_i C_p)^{1/2}$ 500 Hz.

Explanatory notes:

For further details, see sheet no. 201, given in the previous (paper) Collection.